

AY25 PROGRAM ASSESSMENT REPORT

v2026.1

Tracking Data

CIM code (e.g., ME-CM)	PHD-ARCE
Program Name	Doctor of Philosophy in Architectural Engineering
Assessment Year	2024-2025
Program Assessment Coordinator	Brent Stephens
Year of Current Assessment Plan	2024-2025
No Assessment This Year per Plan	☐
LO(s) Assessed	3
Presented at program authority meeting date: <i>Please attach minutes</i>	n/a

Methodology

The objective of the Doctor of Philosophy in Architectural Engineering program is to provide an avenue for the advanced study of the design, construction, and operation of buildings and their systems. Students can specialize in any relevant aspect of Architectural Engineering in which CAEE faculty members have expertise, which includes: building energy and environmental systems, structures, or construction engineering and management. Through state-of-the-art and original research, students are expected to achieve a high level of mastery in the study of buildings and their systems, which should be demonstrated by successful defense of their dissertation and the publication of novel research results in peer-reviewed academic journals. Students who earn a Doctor of Philosophy in Architectural Engineering will be prepared for careers in both academia and industrial research and development. The program is intended for students with bachelor's or master's degrees in architectural engineering, civil engineering, mechanical engineering, or other related disciplines. The doctoral program generally consists of at least two complete years of academic preparation, followed by at least one year of full-time research in residence at the university. The coursework must include core architectural engineering courses and there is flexibility in selecting electives.

To be admitted to candidacy, students must pass a qualifying examination, which involves an oral presentation of two research papers selected by the student's adviser. The qualifying examination is administered by a research committee approved by the chairperson. The exam is diagnostic in nature. The results of the exam will determine the student's potential for success in the Ph.D. program. The department may waive this requirement for students who hold an M.S. degree from Illinois Institute of Technology in the same field. This examination should be completed within three semesters of entry into the program.

The next step in the process is the comprehensive examination, which is an oral examination that is administered by a research committee approved by the chairperson. The candidate presents the research proposal and answers questions of a general professional nature. The research project must be in harmony with the interests of the faculty and with the facilities of the department. The candidate should pass the comprehensive examination at least one year prior to the date of graduation.

Although doctoral research can begin after admission to the Ph.D. program, the major portion of the research should take place after the comprehensive examination is passed and the research proposal is approved by the research committee. Research will be conducted under the supervision of a full-time faculty member and students should work to involve all members of their research committee.

The preliminary thesis draft must meet the approval of all members of the research committee. An oral examination in defense of the thesis is given as an open university seminar. The thesis defense must meet with the approval of the research committee; if it does not, the committee has the authority to determine whether or not to grant a re-examination.

Learning objectives for this program are for students to:

1. Demonstrate mastery in an area of specialization within architectural engineering, with deep knowledge exceeding the fundamentals gained in undergraduate and master's level work.
2. Demonstrate the skills necessary to engage in and conduct original research.
3. Demonstrate the ability to effectively communicate technical information (oral, written, and visual).

The program uses student performance in the qualifying exam to assess learning objectives #1 and performance in the comprehensive exam (dissertation proposal) to assess learning objectives #2 and #3.

In 2020, the CAEE Department initiated new rubrics for assessing both qualifying and comprehensive exams. The rubrics are shown in the appendix of this document.

Results

Data tables and Findings

Learning Objective #1

In Fall 2024, there were 6 students enrolled in the program, each at different stages of progress. Given the size of the program (the PhD program will always be relatively small numbers of students) and the long duration required to complete the program, in this assessment year, data from Fall 2020 to Summer 2025 are used to assessment attainment of learning objectives. A total of 6 students have taken the qualifying exam a total of 7 times in the program since 2020, including 1 in 2021, 1 in 2022, 3 in 2023, 1 again in 2024 (the 1 in 2024 was a repeated attempt), and 1 in 2025 prior to the start of the Fall 2025 semester. Tables 1 and 2 summarize qualifying exams taken in this time period. Only the 2025 qualifying exam taker is from this most recent academic year.

Table 1. Summary of qualifying exams taken this cycle

Student	Date	1 st or 2 nd time?	Result	Chair	Overall Score (out of 21)
1	5/20/2021	1 st	Pass	Heidarinejad	15 (71%)
2	9/30/2022	1 st	Pass	Stephens	12 (57%)
3	2/24/2023	1 st	Fail	Stephens	9.5 (45%)
	8/21/2024	2 nd	Fail	Stephens	11.5 (55%)
4	6/20/2023	1 st	Pass	Stephens	16.5 (79%)
5	7/27/2023	1 st	Pass	Heidarinejad	20 (95%)
6	8/7/2025	1 st	Pass	Heidarinejad	19 (90%)

Table 2. Evaluation of each criterion for each qualifying exam in this cycle

Date	Motiv.	Lit review	Methods/Res.	Fundam.	Oral Comm.	Visual Comm.	Q&A	Total
5/20/2021	2	2	2	2	1	3	2	15
9/30/2022	1.5	1.5	2	1.5	1.5	2	2	12
2/24/2023	1	1	1.5	1.5	1.5	2	1	9.5
8/21/2024	2	2	1.5	1.5	1.5	2	1	11.5
6/20/2023	2	2	2.5	3	2	2.5	2.5	16.5
7/27/2023	3	2.5	2.5	3	3	3	3	20
8/7/2025	2.5	2	3	3	3	2.5	3	19

In this cycle since 2020, 5 of the 6 students who took the qualifying exam passed on their first attempt with an average score of 16.5 out of 21, while the sixth student failed twice. While the evaluation rubric scores are not directly used in determining pass or fail, they serve as a quantitative measure of performance for the purposes of assessment of learning objective #1. Three of the four students who passed the exam on the first attempt had 79% or higher overall scores; the two lower passing scores of 15 (71%) and 12 (57%) highlighted middling scores in motivation, literature review, fundamental knowledge, and oral communication. Most specific to LO #1, 3 of the 5 passing students demonstrated a perfect score on knowledge of fundamental concepts, while the other passing students demonstrated middling scores. Overall scores of <12, 12-18, and ≥18 on the qualifying exam assessment are used to define qualitative assessments of ‘does not meet expectations’, ‘meets expectations’, and ‘proficient’, respectively.

The one student who failed the qualifying exam twice was reevaluated and recommended to switch to the PhD program in Architecture at Illinois Tech; this process was completed in Fall 2024. Upon re-review of the student’s original application, despite having a bachelor’s degree in architectural engineering, it became apparent that architectural engineering at the international school the student came from was much closer to an architecture degree than engineering, which has flagged a revision to our application review processes to be more critical of student transcripts from such programs. Another change to our processes that these performances and experiences have led to is clearer articulation on the part of the research supervisors to clarify for students, prior to taking the exam, more details about how the exam will test fundamental knowledge.

Learning Objectives #2 and #3

A total of 5 students have taken the comprehensive exam in the program since 2020, including 1 in 2021, 2 in 2022, and 2 in 2024. Tables 3 and 4 summarize qualifying exams taken in this time period.

Table 3. Summary of comprehensive exams taken this cycle

Student	Date	1 st or 2 nd time?	Result	Chair	Overall Score (out of 24)
A	12/6/2021	1 st	Pass	Stephens	20 (83%)
B	5/9/2022	1 st	Pass	Heidarinejad	18 (75%)
C	12/5/2022	1 st	Pass	Heidarinejad	20 (83%)
D	10/7/2024	1 st	Pass	Heidarinejad	22.5 (94%)
E	12/3/2024	1 st	Pass	Stephens	15 (63%)

Table 4. Evaluation of each criterion for each comprehensive exam in this cycle

Date	Motiv.	Lit review	Methods/Hyp.	Fundam.	Written	Oral	Visual	Q&A	Total
12/6/2021	3	3	2	2	3	2	3	2	20
5/9/2022	2	2	2	3	3	2	2	2	18
12/5/2022	3	3	2	2	2	2	3	3	20
10/7/2024	3	2.5	2.5	2.5	3	3	3	3	22.5
12/3/2024	2	2	1	2	2.5	2	2	1.5	15

All 5 students passed their comprehensive exams on their first try, with 3 out of 5 students scoring 20 or more out of 24 total possible points. Scores of <14, 14-20, and ≥ 20 on the overall comprehensive exam assessment are used to define qualitative assessments of 'does not meet expectations', 'meets expectations', and 'proficient', respectively, for learning objective #2. All 5 students thus met or exceeded expectations in demonstrating the skills necessary to engage in and conduct original research (LO #2)

Scores of <2, 2-2.75, and ≥ 2.75 on the average of written, oral, and visual communications assessments are used to define qualitative assessments of 'does not meet expectations', 'meets expectations', and 'proficient', respectively, for learning objective #3. All 5 students also met or exceeded expectations in demonstrating the ability to effectively communicate technical information (oral, written, and visual) (LO #3).

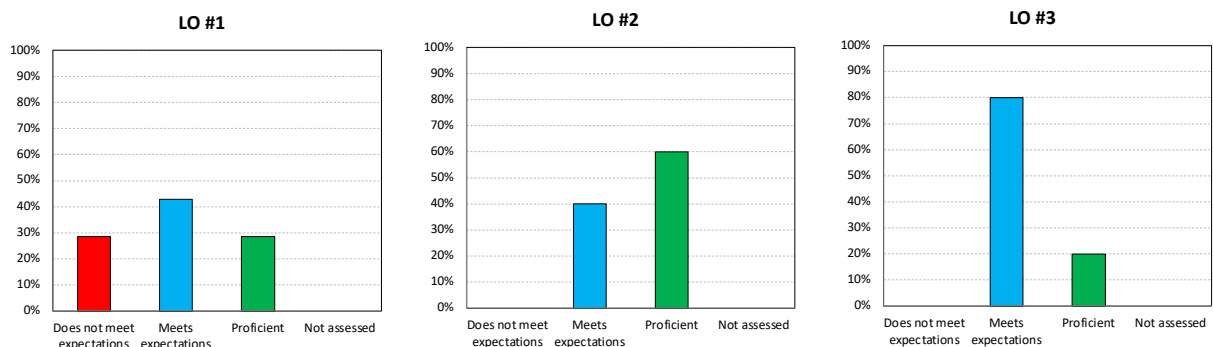


Figure 1. Distribution of proficiency levels for LOs #1, #2, and #3 for this cycle.

Prior CI plan impact

In the previous assessment report, it was noted that student performance on the qualifying exams in the program over the last few years have demonstrated generally sufficient preparation among students (3 of 4 meeting expectations in LO #1), with the lone exception leading to a revision in our application review process to ensure that students are coming in with appropriate backgrounds or are required to take undergraduate prerequisite courses in subjects in which they are lacking. We have continued to implement that process. Student performance on the comprehensive exams in the program over the last few years had also demonstrated good preparation among students in LOs #2 and #3, suggesting that the qualifying exam is working as the primary point of assessment for candidacy and that once admitted to candidacy, students are succeeding in meeting program LOs.

We have also worked to improve our assessment data collection and reporting for this cycle and beyond.

This cycle's CI plan

1. We will continue to use our aggregation approach given the small number of students taking either exam in any given year.
2. We will continue to use the new assessment data analysis and reporting templates.
3. We will continue to implement past CI plans in terms of evaluating applicants for the program.

Appendices – See attached for qualifying and comprehensive exam rubrics

IIT CAEE – PhD Qualifying Exam Evaluation Rubric

Instructions: The student receives two papers from the literature, selected by the student's advisor. The papers may or may not be directly related to their proposed research topic. Students have at least two weeks to prepare a critical review of those papers and present the research in those papers in depth in an oral presentation. This format is used as a launch point for deeper interrogation into the student's knowledge of the fundamentals in their area of specialization. The qualifying exam committee will then complete the following evaluation rubric for each candidate taking the qualifying exam using a consensus approach.

Student: _____		Program (CE, ARCE, ENVE): _____		Exam Date: _____	
Criterion	Below Expectations (1)	Meets Expectations (2)	Exceeds Expectations (3)	Score (1-3) & Comments	
1. Motivation for the work in the two papers	The reasons for the work are not covered or only minimally covered.	Big picture presented. Reasons for research question laid out.	Motivation is clear and documentation and/or data is used to show the importance and need for the work.		
2. Literature review in the two papers	Limited background. No context. Limited or no signs of critical thinking regarding other work.	Relevant papers and work are cited and understood.	Relevant papers and work are cited, understood, and evaluated critically. Review goes deeper than just the two assigned papers.		
3. Methods and results in the two papers	Methods and results are not well understood or evaluated critically.	Methods and results are understood and evaluated. Limitations are identified, with some assistance needed.	Methods and results are well understood and critically evaluated. Limitations are well defined.		
4. Demonstration of knowledge of fundamental concepts in the student's area of specialization	Core material is not understood well.	Core material is referenced, and relevant parts are used in the exam.	Core material is referenced and used to demonstrate limitations of the work or propose new and potentially important research questions.		
5. Oral communication by the student	Limited command of language and vocabulary. Speaker cannot be heard clearly.	Command of language and vocabulary, with some clarification needed. Speaker projects well.	Speaker projects well and demonstrates excellent command of language and vocabulary.		
6. Visual communication by the student	Slides hard to read. Organization poor. Grammatical or spelling mistakes.	Slides are clear. Presentation is organized.	Slides are superb. Presentation is superb. Speaker is on par with speakers at national conferences.		
7. Responses to questions by the student	Does not understand questions. Is not able to answer questions.	Understands questions. Answers questions, potentially with clarifications.	Understands and responds to questions as well as gives context to larger issues around questions.		
Exam Result (Pass/Fail): _____				Overall Score: _____ (out of 21)	

Additional Comments (Optional):

Committee Signatures:

IIT CAEE – PhD Comprehensive Exam Evaluation Rubric

Instructions: The student submits a written research proposal and then presents their dissertation proposal orally. The committee evaluates the proposal on its technical foundation, novelty, feasibility, likelihood of success, and clarity. The comprehensive exam can also include general questions about any material relevant to the research. The comprehensive exam committee will then complete the following evaluation rubric for each candidate taking the comprehensive exam using a consensus approach.

Student: _____		Program (CE, ARCE, ENVE): _____		Exam Date: _____	
Criterion	Below Expectations (1)	Meets Expectations (2)	Exceeds Expectations (3)	Score (1-3) & Comments	
1. Motivation for the work and demonstration of background work	The reasons for the work are not covered or only minimally covered. Minimal background research to date.	Big picture presented. Reasons for research question laid out. Some background research to date, with conference paper published to demonstrate.	Motivation is clear and documentation and/or data is used to show the importance and need for the work. Successful background research completed to date, with peer-reviewed publication(s) to demonstrate.		
2. Literature review	Limited background. No context. Limited or no signs of critical thinking regarding other work.	Relevant papers and work are cited and understood. Review goes beyond the two assigned papers. Some command of existing literature is shown.	Relevant papers and work are cited, understood, and evaluated critically. Full command of existing literature is demonstrated.		
3. Methods and hypotheses	Methods are not well understood or unlikely to yield success. Testable hypotheses not developed.	Methods are understood and evaluated. Limitations are identified, with some assistance needed. Testable hypotheses are developed and have a potential for novel outcomes.	Methods are well understood and critically evaluated. Limitations are well defined. Testable hypotheses are well developed and highly novel.		
4. Fundamental concepts	Proposal does not build on an understanding of fundamental concepts.	Proposal builds on an understanding of some fundamental concepts.	Proposal builds on a complete understanding of fundamental concepts and mastery is demonstrated.		
5. Written communication	Limited command of language and vocabulary. Grammatical mistakes. Poor organization. Figures are poor.	Command of language and vocabulary. Minimal grammatical mistakes. Proposal is organized. Figures are readable.	Full command of language and vocabulary. No grammatical mistakes. Proposal is well organized and complete. Figures are publication quality.		
6. Oral communication	Limited command of language and vocabulary. Speaker cannot be heard clearly.	Command of language and vocabulary, with some clarification needed. Speaker projects well.	Speaker projects well and demonstrates excellent command of language and vocabulary.		
7. Visual communication	Slides hard to read. Organization poor. Grammatical or spelling mistakes.	Slides are clear. Presentation is organized.	Slides are superb. Presentation is superb. Speaker is on par with speakers at national conferences.		
8. Responses to questions	Does not understand questions. Is not able to answer questions.	Understands questions. Answers questions, potentially with clarifications.	Understands and responds to questions as well as gives context to larger issues around questions.		
Exam Result (Pass/Fail): _____		Overall Score: _____ (out of 24)			

Additional Comments (Optional):

Committee Signatures: